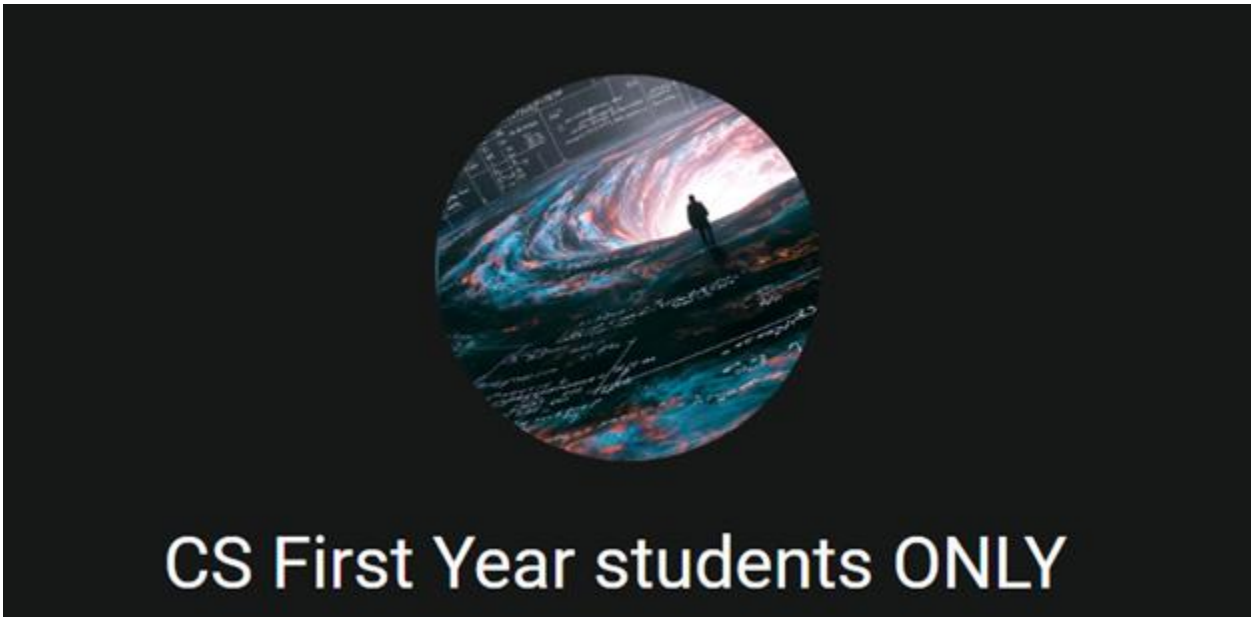
Midterm Exam For Second Semester (2024/2025)



Leader's :

1 - Jannah Ahmed

2 - Kareem Yousry

Ministry of High Education
Obour High Institute for Management & Informatics

معاهد العبور
Obour Institutes

Midterm Exam for second Semester (2024/2025)

Course Name: Linear Algebra	Course Code: BS 102 Bylaw: 2024
Year: First	Department: Computer Science
Date of Exam: Saturday 25/3/2025	Time Allowed: 1 hour Total Points: 20
Course Instructor: Dr. Dina Ezzat Mahmoud	No. of questions: (3) No. of pages: (1)

Exam instructions

- **Make sure that the question form is identical to the answer sheet form**
- **The answer sheet must be in good condition from the moment received until it is delivered after taking the exam**
- **Do not bend the answer sheet or lean on it while answering**

Answer the following questions:

Question One: [6 points] (ILOs: a1, b1, c2)

Choose the correct answer from (A), (B), (C) or (D), to complete the sentence:

1- **Which one of the following is elementary matrix**

a- $\begin{bmatrix} 1 & 0 \\ 0 & 2 \\ 0 & a \end{bmatrix}$ b- $\begin{bmatrix} 4 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 4 \end{bmatrix}$ c- $\begin{bmatrix} 4 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ d- $\begin{bmatrix} 1 & 0 \\ 2 & 4 \end{bmatrix}$

2- **When multiplying 3 by the matrix** $\begin{bmatrix} 1 & -2 & 2 \\ 3 & 4 & 1 \\ 2 & -1 & 1 \end{bmatrix}$

a- $\begin{bmatrix} 3 & -6 & 6 \\ 9 & 12 & 3 \\ 6 & -3 & 3 \end{bmatrix}$ b- $\begin{bmatrix} 3 & -6 & 6 \\ 3 & 4 & 1 \\ 2 & -1 & 1 \end{bmatrix}$ c- $\begin{bmatrix} 1 & -2 & 2 \\ 9 & 12 & 3 \\ 2 & -1 & 1 \end{bmatrix}$ d- $\begin{bmatrix} 1 & -2 & 2 \\ 3 & 4 & 1 \\ 6 & -3 & 3 \end{bmatrix}$

3- **Which one of the following is non-linear equation**

a- $3x + y = 5$ b- $3x + \pi y + z = 0$ c- $\sqrt{2x} + y = 0$ d- $x^2 - y = 6$

Question Two: [6 points] (ILOs: a1, b1, c2)

Use the matrix $A = \begin{bmatrix} 1 & 2 & -1 \\ 3 & 1 & 5 \\ 1 & 2 & 4 \end{bmatrix}$ to encode a message "Meet Me Sunday".

Question Three: [8 points] (ILOs: a1, b1, c2)

Use Gauss-Jordan elimination to find the inverse of the matrix $B = \begin{bmatrix} 1 & 2 & 2 \\ 3 & 7 & 9 \\ -1 & -4 & -7 \end{bmatrix}$

End of Questions
Good Luck,,,

Examiner:
Dr. Dina Ezzat Mahmoud